

**SOLUTION**

$$d = 50 \text{ mm} = 50 \times 10^{-3} \text{ m}, I_1 = I_2 = 200 \text{ A}$$

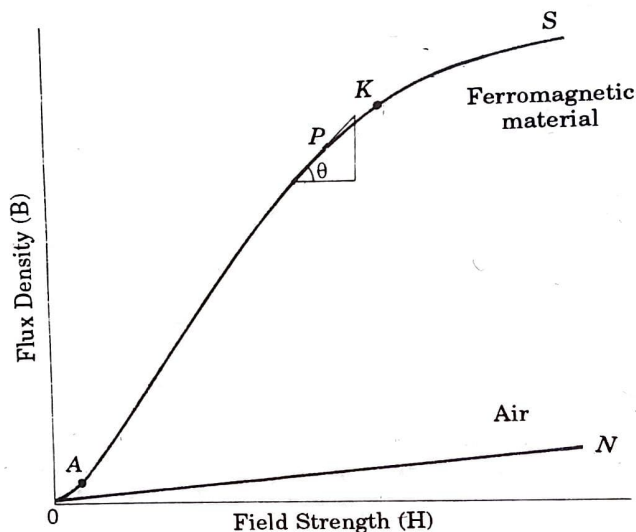
Force on each conductor per metre length

$$F_c = 2 \times 10^{-7} \frac{I_1 I_2}{d} = 2 \times 10^{-7} \times \frac{200 \times 200}{50 \times 10^{-3}} = 0.16 \text{ N}$$

Since the conductors carry the currents in the same direction, therefore, there is a force of attraction between them.

**7.8 MAGNETISATION CURVE OR B-H CURVE**

The graph between the flux density  $B$  and field intensity  $H$  of a magnetic material is called the *magnetisation curve* or *B-H curve*. In air or a non-magnetic material, the flux density  $B$  is proportional to magnetic field intensity  $H$ . Therefore the graph of  $B$  versus  $H$  is a straight line through the origin as shown in Fig. 7.2.



**Fig. 7.2.** B-H curves for air and a ferromagnetic material.

A typical  $B-H$  curve for a ferromagnetic material is also shown in Fig. 7.2. Except for a small portion near the origin, the curve is linear from A to K. Over this portion the flux density  $B$  is proportional to the field intensity  $H$  and  $\mu_r$  is constant. For higher values of  $H$ , there is no further increase in  $B$  with  $H$ . The curve becomes almost horizontal after the point K. The material is then said to be saturated and the point K is called the point of saturation for the material.

The slope of the  $B-H$  curve at any point  $P$  is given by

$$\tan \theta = \frac{B}{H}$$

$$B = \mu_0 \mu_r H; \quad \mu_r = \frac{1}{\mu_0} \cdot \frac{B}{H} = \frac{1}{\mu_0} \tan \theta$$

Thus,  $\mu_r$  is proportional to the slope of the  $B-H$  curve at any point. Starting from a definite value at the origin, the slope increases as  $B$  increases until it becomes a maximum. It then gradually

decreases as  $B$  increases further. The slope becomes almost zero in the saturation region when the curve becomes horizontal. The  $B-H$  curve shows that the permeability  $\mu_r$  of a magnetic material changes with the flux density  $B$ .

## 7.9 FERROMAGNETIC MATERIALS

The magnetic behaviour of materials can be explained on the basis of Bohr's theory of atomic structure. According to this theory the electrons revolve around the nucleus in fixed orbits. Since these electrons are in motion they constitute electric currents. These currents produce magnetic fields called *orbital magnetic fields*. In addition to this, each electron *spins* on its axis as it revolves in an orbit. The spin of an electron is similar to the spin of the earth. A spinning electron is a charge in motion and, therefore, constitutes an electric current. This current produces a magnetic field called the *spin magnetic field*.

For an individual atom these fields are very weak. Strong magnetic fields can be produced if the atoms are grouped in a material in such a way that their orbital and spin fields reinforce one another. Such materials are called ferromagnetic materials. In such materials there is an appreciable interaction between neighbouring atoms. Atoms do not act singly but in groups called domains. Each domain contains between  $10^9$  and  $10^{15}$  atoms. All the atoms in any one domain have their magnetic axes in the same direction and are aligned. Each domain is a region of intense magnetisation. However, the domains are randomly oriented in an unmagnetised specimen so that their magnetic effects cancel one another. This random arrangement of domains is shown in Fig. 7.3.

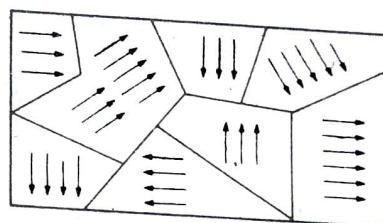


Fig. 7.3 Random arrangement of domains.

If an external magnetic field is applied, the atoms align themselves with the applied field. Consequently, the resultant magnetic flux density *inside* the material is much greater than the flux density of the applied field. Examples of ferromagnetic materials are iron, nickel, cobalt and their alloys.

## 7.10 EXPLANATION OF THE SHAPE OF $B-H$ CURVE

The shape of  $B-H$  curve can be explained on the basis of *domain* theory. Initially the specimen is magnetically neutral because the domains are randomly oriented so that their magnetic effects cancel one another. If an external magnetic field is applied, the domains having their magnetic axes close to the direction of the applied field grow in size at the expense of their neighbouring domains. This growth is reversible and if the applied field is removed, the flux also becomes zero. The portion OA (shown exaggerated) on the  $B-H$  curve is due to this initial alignment. As  $H$  is increased further, more domains align themselves with the field and grow in size. This growth is reversible and if the applied field is removed, the flux will still be there due to the aligned domains. The building of domains at a steady rate is represented by the linear portion AK of the  $B-H$  curve. At higher values of  $H$ , rotation of the remaining domains takes place. When most of the domains are aligned the material is said to be *magnetically saturated*. It is shown by the portion KS of the  $B-H$  curve. The point K where the saturation begins is called the *knee* of the curve.

## LAWS GOVERNING MAGNETIC CIRCUITS

It is sometimes convenient to compare magnetic circuits with dc electric circuits for simplifying the calculations of magnetic circuits. There is a close mathematical analogy between magnetic and dc circuits. For a dc resistive circuit, the Ohm's law is given by



$$\text{current, } I = \frac{\text{e.m.f.}}{\text{resultance}} = \frac{E}{R}$$

For a magnetic circuit,

$$\text{flux, } \Phi = \frac{\text{m.m.f.}}{\text{reluctance}} = \frac{F}{S}$$

If we compare these two relations, we see that the flux is analogous to current, m.m.f. is analogous to e.m.f., and reluctance is analogous to resistance except that reluctance is not an energy-loss component.

Also,

$$R = \rho \frac{l}{a}$$

and

$$S = \frac{1}{\mu_0 \mu_r} \cdot \frac{l}{a}$$

These two equations are also similar. Table 7.1 gives a comparison of electric and magnetic quantities.

Table 7.1

| Magnetic circuit         |           |      | DC resistive circuit     |        |          |
|--------------------------|-----------|------|--------------------------|--------|----------|
| Quantity                 | Symbol    | Unit | Quantity                 | Symbol | Unit     |
| m.m.f.                   | $F$       | A    | e.m.f.                   | $E$    | V        |
| Magnetic field intensity | $H$       | A/m  | Electric field intensity | $E$    | V/m      |
| Magnetic flux            | $\Phi$    | Wb   | Current                  | $I$    | A        |
| Reluctance               | $S$       | A/Wb | Resistance               | $R$    | $\Omega$ |
| Permeance                | $\Lambda$ | Wb/A | Conductance              | $G$    | S        |

$$F = S\Phi$$

$$S = \frac{l}{\mu_0 \mu_r a}$$

$$E = RI$$

$$R = \rho \frac{l}{a}$$

We have seen that analog of resistance is *reluctance*, the analog of electromotive force (emf) is *magnetomotive force* ( $F$ ), and the magnetic analog of current is the *flux*,  $\Phi$ . Thus magnetic Ohm's law may be written as

$$\text{flux} = \frac{\text{mmf}}{\text{reluctance}}$$

$$\Phi = \frac{F}{S}$$

By analogy, the laws of resistances in series and parallel also hold for reluctances.

The equivalent reluctance of a number of reluctances in series is equal to the sum of individual reluctances :

$$S_{eq} = S_1 + S_2 + S_3 + \dots + S_n$$

In parallel magnetic circuits, the same mmf is applied to each of the parallel paths and the total flux divides between the paths in inverse proportion to their reluctances.

$$\Phi = \Phi_1 + \Phi_2 + \Phi_3 + \dots + \Phi_n$$

The equivalent reluctance  $S_{eq}$  of a number of reluctances in parallel is given by

$$\frac{1}{S_{eq}} = \frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \dots + \frac{1}{S_n}$$

Permeances in series and parallel obey the same rules as electrical conductances.

The equivalent of the two Kirchhoff's laws for the magnetic circuit are as follows :

- The sum of fluxes entering a junction in a magnetic circuit is equal to the sum of the fluxes leaving the junction. This is analogous to KCL for electric circuits.
- In a closed path of a magnetic circuit the algebraic sum of mmfs (magnetic potential drops) required to force the flux through the elements must be equal to the net ampere turns of excitation. This is analogous to KVL for the electric circuits. Thus the sum of magnetic potentials around any closed path is equal to zero.

Scalar magnetic potential

$$F = \int \mathbf{H} \cdot d\mathbf{l}$$

In order to apply these laws, the circuit is divided into a number of elements selected in such a way that the flux density is approximately the same throughout the element. The length of each element corresponds to its mean path.

Because of the analogy between the magnetic circuits and dc resistive circuits, it is possible to use techniques to solve magnetic circuits similar to those used for dc resistive circuits. The equivalent circuits are drawn and Ohm's law, KCL, and KVL equations are written for these equivalent dc resistance circuits. The corresponding equations for magnetic quantities are written by analogy.

Some of the commonly used magnetic circuits are shown in Fig. 7.4. The equivalent electric circuits and their analogs are also shown. Circuit equations are written using Ohm's law, KCL and KVL, and the corresponding equations for magnetic circuits are also written by analogy.

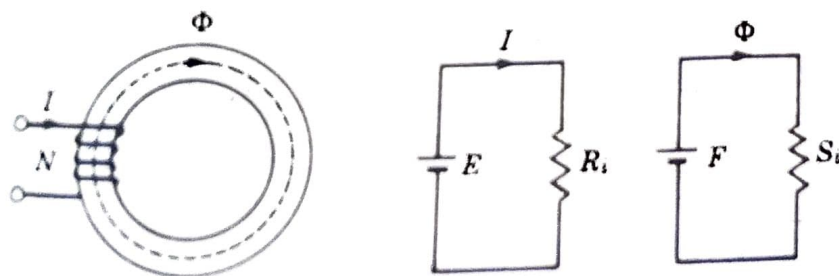


Fig. 7.4 (a)

$$E = R_i I, \quad F = S_i \Phi = H_i l_i$$

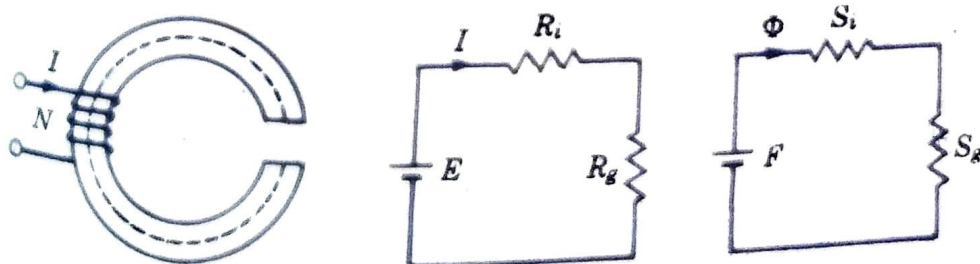


Fig. 7.4 (b)

$$E = (R_i + R_g) I$$

$$\begin{aligned} F &= (S_i + S_g) \Phi \\ &= F_i + F_g = H_i l_i + H_g l_g \end{aligned}$$



Total m.m.f.

$$F = F_a + F_i = 2026 + 5093 = 7119 \text{ A}$$

Magnetising current

$$I = \frac{\text{m.m.f.}}{\text{number of turns}} = \frac{7119}{2000} = 3.56 \text{ A}$$

## 7.16 SOLUTION OF MAGNETIC CIRCUITS FROM $B$ - $H$ CURVES

In practice,  $B$ - $H$  curve is used for the solution of magnetic circuits. The following procedure is used

- (1) The value of  $B$  is determined for each part of the circuit.
- (2) The corresponding value of  $H$  is obtained from the  $B$ - $H$  curve for the particular material.
- (3) The magnetic path length is computed.
- (4) The value of  $H$  obtained in step (2) is multiplied with the magnetic path length to get m.m.f. for the particular part of the circuit.

### EXAMPLE 7.7

A magnetic circuit comprises three parts,  $a$ ,  $b$  and  $c$  in series, each of which has uniform cross-sectional area. Part  $a$  has a length of 80 mm and cross-sectional area  $50 \text{ mm}^2$ . Part  $b$  has a length of 70 mm and cross-sectional area  $80 \text{ mm}^2$ . Part  $c$  is an air-gap 0.5 mm in length and of cross-sectional area  $60 \text{ mm}^2$ . Neglecting magnetic leakage and fringing, determine the current necessary in a coil of 4000 turns wound on part  $b$  to produce in the airgap a flux density of 0.7 T. The magnetic characteristic for parts  $a$  and  $b$  is given by :

|           |     |     |     |     |     |      |
|-----------|-----|-----|-----|-----|-----|------|
| $H$ (A/m) | 100 | 210 | 340 | 500 | 800 | 1500 |
| $B$ (T)   | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | 1.2  |

### SOLUTION

The  $B$ - $H$  curve is plotted from the given data. It is shown in Fig. 7.10. Since the given circuit is a series circuit, the flux is the same in every part of the circuit. That is,

$$\text{flux in part } a = \text{flux in part } b = \text{flux in part } c$$

Flux in the airgap (part  $c$ ).

$$\Phi_c = \text{flux density in the airgap} \times \text{area of the airgap}$$

$$\text{Area of the airgap} = 60 \text{ mm}^2 = 60(10^{-3} \text{ m})^2 = 60 \times 10^{-6} \text{ m}^2$$

$$\begin{aligned} \therefore \Phi_c &= B_c \times a_c \\ &= 0.7 \times 60 \times 10^{-6} = 42 \times 10^{-6} \text{ Wb} \end{aligned}$$

Calculation of m.m.f. for part  $a$

$$l_a = 80 \text{ mm} = 80 \times 10^{-3} \text{ m}$$

$$a_a = 50 \text{ mm}^2 = 50 \times 10^{-6} \text{ m}^2$$

$$B_a = \frac{\Phi_a}{a_a} = \frac{\Phi_c}{a_a} = \frac{42 \times 10^{-6}}{50 \times 10^{-6}} = 0.84 \text{ T}$$

From the  $B$ - $H$  curve, the value of magnetic field strength for this value of flux density is  $H_a = 540 \text{ A/m}$ .

$$\text{m.m.f. required for part } a \quad F_a = H_a l_a = 540 \times 80 \times 10^{-3} = 43.2 \text{ A}$$

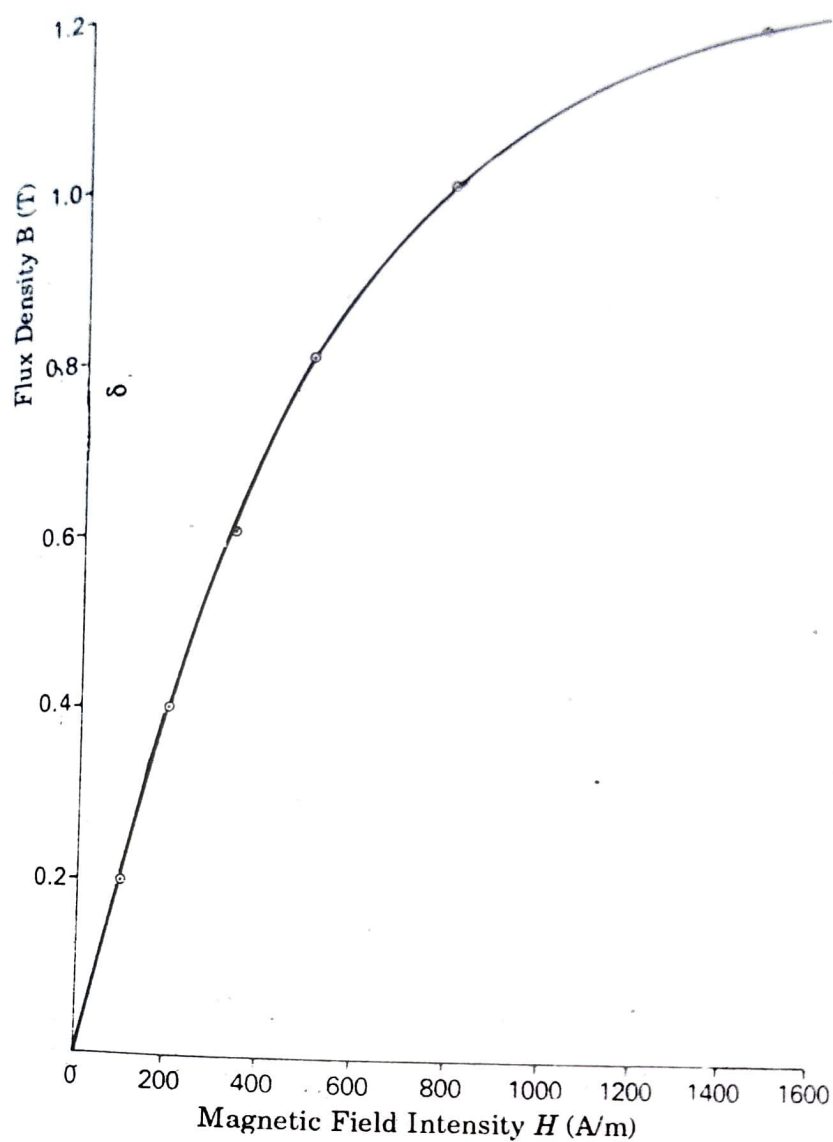


Fig 7.10

Calculation of *m.m.f.* for part *b*.

$$l_b = 70 \text{ mm} = 70 \times 10^{-3} \text{ m}$$

$$a_b = 80 \text{ mm}^2 = 80 \times 10^{-6} \text{ m}^2$$

$$B_b = \frac{\Phi_b}{a_b} = \frac{\Phi_c}{a_b} = \frac{42 \times 10^{-6}}{80 \times 10^{-6}} = 0.525 \text{ T}$$

From the *B-H* curve, the value of magnetic field strength for this value of flux density,

$$H_b = 280 \text{ A/m}$$

*m.m.f.* required for part *b*

$$F_b = H_b l_b = 280 \times 70 \times 10^{-3} = 19.6 \text{ A}$$

Calculation of *m.m.f.* for part *c* (airgap)

$$l_c = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$$

$$a_c = 60 \text{ mm}^2 = 60 \times 10^{-6} \text{ m}^2$$

$$B_c = 0.7 \text{ T}$$

$$H_c = \frac{B_c}{\mu_0} = \frac{0.7}{4\pi \times 10^{-7}} = 5.57 \times 10^5 \text{ A/m}$$

m.m.f required for airgap

$$F_c = H_c l_c = 5.57 \times 10^5 \times 0.5 \times 10^{-3} = 278.5 \text{ A}$$

Total m.m.f. of the circuit

$$\begin{aligned} F &= F_a + F_b + F_c \\ &= 43.2 + 19.6 + 278.5 = 341.3 \text{ A} \end{aligned}$$

Exciting current

$$I = \frac{F}{N} = \frac{341.3}{4000} = 0.0853 \text{ A}$$

### EXAMPLE 7.8

Determine the total mmf acting in the magnetic circuit shown in Fig. 7.11.

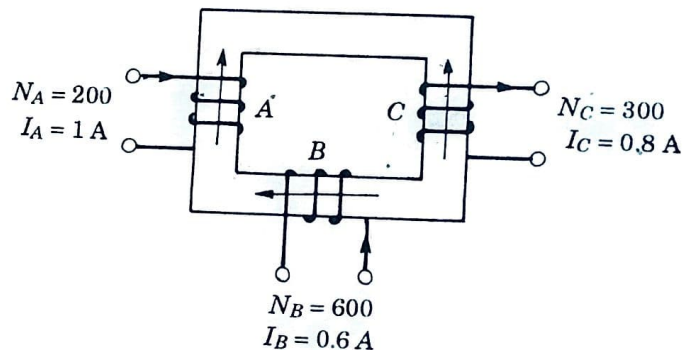


Fig. 7.11

### SOLUTION

Total mmf acting in the circuit

$$\begin{aligned} &= I_A N_A + I_B N_B - I_C N_C \\ &= 1 \times 200 + 0.6 \times 600 - 0.8 \times 300 = 320 \text{ A} \end{aligned}$$

### EXAMPLE 7.9

A ring of cast steel has an external diameter of 24 cm and a square cross-section of 3 cm side. Inside and across the ring, an ordinary steel bar 18 cm × 3 cm × 0.4 cm is fitted with negligible gap. Calculate the number of ampere turns required to be applied to one half of the ring to produce a flux density of 1 T in the other half. Neglect leakage. The  $B$ - $H$  characteristics are as follows :

|                 | For cast steel |      |      |     | For ordinary steel |      |
|-----------------|----------------|------|------|-----|--------------------|------|
| $B(\text{T})$   | 1.0            | 1.1  | 1.2  | 1.2 | 1.4                | 1.45 |
| $H(\text{A/m})$ | 900            | 1020 | 1220 | 590 | 1200               | 1650 |

### SOLUTION

The magnetic circuit is shown in Fig. 7.12 (a). The equivalent electric circuit and its magnetic analog circuit are shown in Fig. 7.12 (b) and Fig. 7.12 (c) respectively.



Flux density through area  $a_1$

$$B = \frac{\Phi}{a_1} = \frac{10.74 \times 10^{-4}}{9 \times 10^{-4}} = 1.1933 \text{ T}$$

From the given  $B - H$  table for cast steel, A/m required to produce a flux density of 1.1933 T is  $H = 1206 \text{ A/m}$

Therefore mmf required for portion (1)

$$F_1 = H_1 l_1 = 1206 \times 0.33 = 397.98 \text{ A}$$

By KVL in the left-hand mesh of Fig. 7.12 (b),

$$E - I_1 R_1 - I_3 R_3 = 0$$

$$E = I_1 R_1 + I_3 R_3$$

By analogy

$$F = \Phi_1 S_1 + \Phi_3 S_3 = F_1 + F_3 = 397.98 + 297 = 694.98 \text{ A}$$

Therefore total mmf required

$$= 694.98 \text{ A}$$

## 7.17 MAGNETIC HYSTERESIS

Consider a coil wound on an iron core which is completely demagnetised. The field intensity in the coil is given by  $H = IN/l$ . The value of  $H$  can be increased or decreased by increasing or decreasing the current through the coil. When the current in the coil is zero,  $H = 0$  and  $B = 0$ . If the current and hence the field strength  $H$  increases gradually, the flux density  $B$  in the core also increases. The relationship between  $B$  and  $H$  is represented by the curve  $OA$  shown in Fig. 7.13. At point  $A$ , the core becomes magnetically saturated and the field strength  $H$  reaches its highest value  $H_m$  corresponding to the maximum value of the magnetising current. At this point the flux density in the core has a maximum value of  $B_m$ .

If  $H$  is gradually reduced to zero, it will be seen that the corresponding values of  $B$  tend to lag behind the changes in  $H$ . This is called *magnetic hysteresis*. The  $B-H$  curve for the decreasing values is  $AR$ . It lies above the curve  $OA$  for increasing values of  $B$  and  $H$ . At point  $B$ , when  $H = 0$ ,  $B$  is not zero but has a value equal to  $OR$ . This shows that when the field strength is reduced to zero the core is not completely demagnetised. There still remains a certain flux, called the *residual flux density* or *remanent flux density*. This remaining value  $H_m$  to zero is called *remanence* of the core material. It is denoted by  $B_r$ . In Fig. 7.13,  $B_r = OR$ . The residual flux density in the core is due to the fact that the domains do not entirely revert to their original disorder. Some domains still remain aligned after  $H$  is reduced to zero.

To demagnetise the core (i.e., to remove the residual flux),  $H$  is reversed by reversing the current in the coil. The flux density falls to zero along the curve  $RC$ . At  $C$ ,  $H = H_c$  and  $B = 0$ . The demagnetising field strength  $H_c$  required to remove the residual flux density is called the *coercive*

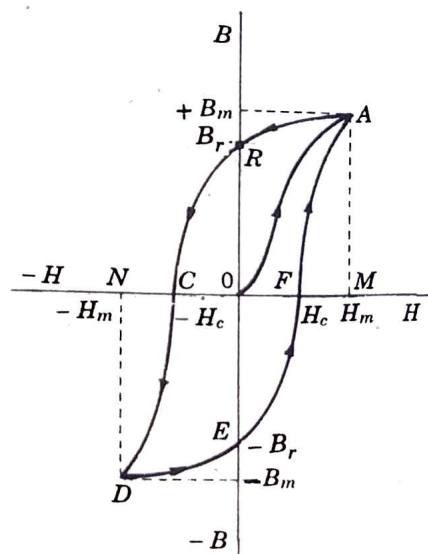


Fig. 7.13. Hysteresis loop.



force. The value of  $H_c$  when the material has been taken to saturation is called the *coercivity* of the material.

If  $H$  is further increased in the reversed direction to  $-H_m$ , the curve follows the path CD. Point D represents saturation in the reversed direction and  $ND = -i$ . If  $H$  is again gradually decreased to zero, the curve follows the path DE. At point E,  $H = 0$ ,  $B = -B_r$ . To remove the residual flux,  $H$  is applied in the original positive direction and the curve follows the path EF. At point F,  $H = H_c$  and  $B = 0$ . If  $H$  is further increased in this direction, the curve follows the path FA. The closed loop ARCDEFA obtained when the magnetic core is subjected to one complete cycle of magnetisation is called the *hysteresis loop*. To take the core through the complete cycle, an alternating magnetic field intensity of peak-to-peak value equal to NM is applied. If this alternating  $H$  is maintained, the core continues to follow similar cycles of magnetisation.

Except at points A and D where  $B$  and  $H$  attain their maximum values, the flux density  $B$  lags behind  $H$  throughout the cycle. For example, at point F when  $B = 0$ ,  $H$  has already reached a positive value  $+H_c$ . At point E, when  $H = 0$ ,  $B$  has a negative value  $-B_r$ . The word *hysteresis* means lagging behind. The curve gets its name from the fact that the flux density (effect) lags being the magnetic field intensity  $H$  (cause).

Hysteresis results from reversals of flux of any value. For a given specimen, a number of hysteresis loops may be obtained, each for a different maximum value of  $H$ . All these loops lie within major or largest hysteresis loops corresponding to the saturation values of  $B$  and  $H$ . A family of such loops is shown in Fig. 7.14. The apexes of all minor loops lie on a curve called the *normal magnetisation curve* shown dotted in the figure.

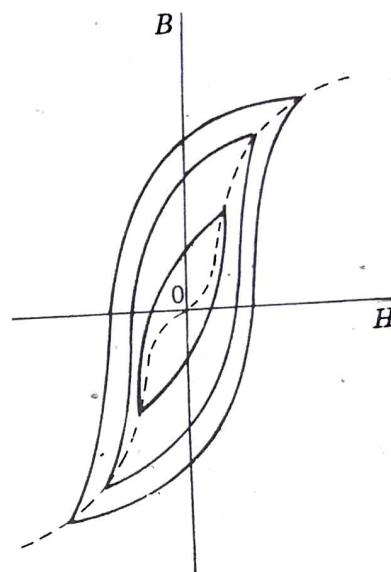


Fig. 7.14. Family of hysteresis loops.

## 7.18 CORE LOSSES OR IRON LOSSES

Time-varying fluxes produce losses in ferromagnetic materials, known as core losses or iron losses. The core losses consist of hysteresis and eddy-current losses. That is  $P_{\text{core}} = P_h + P_e$ . These losses do not occur in ferromagnetic cores that carry flux which does not vary with time. Both hysteresis loss and eddy-current loss produce heat in a magnetic circuit. Within an electric machine hysteresis and eddy-current losses occur simultaneously.

## 7.19 HYSTERESIS LOSS

When a material is magnetised, its domains are aligned in the direction of the applied field. If the material is magnetised in the reverse direction, the domains swing around to align themselves in the opposite direction. If there is a cyclic reversal of the applied field, the domains rotate to and fro. In this process, there is a loss of power called hysteresis loss. This loss appears in the form of heat and raises the temperature of the magnetic material.

Consider the coil-core assembly shown in Fig. 7.15 (a). The hysteresis loops in Fig. 7.14 are obtained by slowly varying a current  $i$  (Fig. 7.15 (b)) of the coil over a cycle. When the current  $i$  is varied through a cycle, during some interval of time energy flows from the source to the coil-core assembly. During some other interval of time energy returns to the source. However, the energy

flowing in is greater than the energy returned. Therefore, during a cycle of variation of  $i$ , there is a net energy flow from the source to the coil-core assembly. This energy loss heats the core. The power loss in the core due to the hysteresis effect is called *hysteresis loss*. It will be proved that the area of the hysteresis loop is proportional to the hysteresis loss.

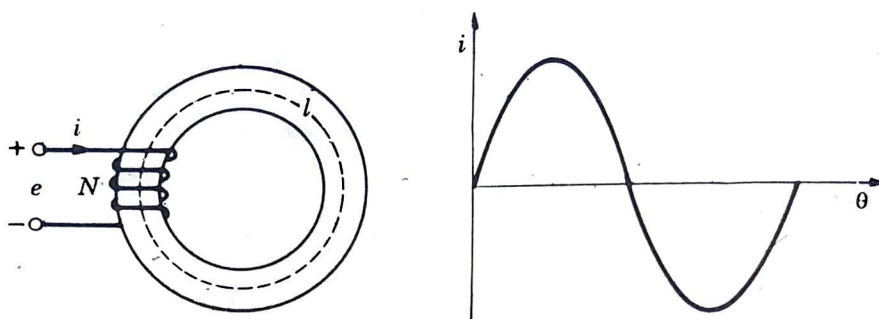


Fig 7.15

Suppose that the coil in Fig. 7.15 (a) has no resistance and the flux in the core is  $\Phi$ .

Let  $l$  = mean length of the core  
 $a$  = area of cross-section of the core  
 $N$  = number of turns of the coil

Since the current changes through the coil, the flux will also change. Hence by Faraday's law, an emf will be induced in the coil given by

$$e = N \frac{d\Phi}{dt}$$

Since  $\Phi = Ba$

$$e = Na \frac{dB}{dt}$$

$$iN = Hl$$

$$i = \frac{Hl}{N}$$

$$\therefore ei = al \frac{HdB}{dt}$$

Energy spent in time  $dt$  is

$$dW = \text{power} \times dt = ei \, dt$$

$$\therefore dW = al \, HdB$$

The total energy transfer  $W$  when  $B$  is increased from a value  $B_1$  to another value  $B_2$  is obtained by integrating  $HdB$  between the limits  $B_1$  to  $B_2$

$$\therefore W = al \int_{B_1}^B HdB$$

But  $al = v_{\text{core}} = \text{volume of the core}$



$$W = v_{\text{core}} \int_{B_1}^{B_2} H dB$$

The integral  $\int_{B_1}^{B_2} H dB$  represents the area shown shaded in Fig. 7.16. The energy transfer for one cycle of magnetisation is

$$W_{\text{cycle}} = v_{\text{core}} \oint H dB$$

$$W_{\text{cycle}} = v_{\text{core}} \times \text{area of the B-H loop} = v_{\text{core}} \times W_h$$

$\oint H dB = \text{area of the B-H loop}$ . If  $f$  is the frequency of variation of the current  $i$ , then  $f$  cycles will be produced per second. Therefore, the power loss in the core due to hysteresis

$$P_h = v_{\text{core}} W_h f$$

It is seen that the hysteresis loss is proportional to the area of the hysteresis loop. It also depends on the volume of the core and it is also directly proportional to the frequency of magnetisation. The hysteresis loss in a magnetic circuit is reduced by using good quality electrical steel for the material. The hysteresis loop for such a material should be narrow.

The armature cores of generator, motors and the cores of transformers are subjected to cyclic variation of flux. Hence there is a continuous hysteresis loss in them resulting in the reduction of efficiency and rise in their temperature. This loss is kept low by using materials having low hysteresis loss. Silicon is added to steel for this purpose. For stationary machines like transformers, silicon content is up to 4% and for rotating machines it is up to 2%.

It is not convenient to find out the area of the hysteresis loop because the B-H curve is non-linear and multivalued and no simple mathematical expression can describe the loop. Several empirical formulae for computing the hysteresis loss  $P_h$  have been developed. The following empirical formula was evolved by Steinmetz and is widely used.

$$P_h = v k_h B_m^n f$$

$v$  = volume of the core

$k_h$  = a constant whose value depends upon the ferromagnetic material

$B_m$  = maximum value of the flux density

$f$  = frequency of variation of the current  $i$

The exponent  $n$  varies in the range 1.5 to 2.5 depending on the material. However, it is important to note that the hysteresis loss varies directly as the frequency for a given material.

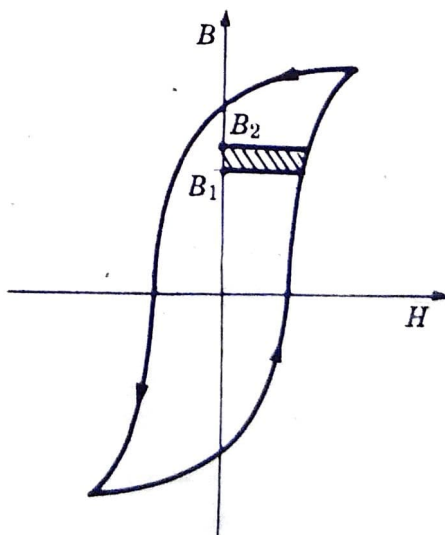


Fig. 7.16

## 7.20 EDDY CURRENT LOSS

Whenever the magnetic flux linking a conducting material circuit changes a voltage is induced in it. Consider a solid iron core (Fig. 7.17 (a)) with which a time varying flux is linking. The core may be considered to be made up of concentric shells. Since iron is a conducting material, each shell may be treated as a closed coil. Each shell links with some portion of the flux. Hence a voltage is induced in each shell. Each induced voltage causes a current to flow in its closed shell. Since these currents do not flow in any clearly defined path, they are called *eddy currents*. Heat is generated due to the flow of eddy currents through the core. The power loss due to eddy currents is called *eddy-current loss*.

The eddy-current loss per unit volume of a magnetic core subjected to a time-varying flux is given by

$$P_e = \frac{1.645}{\rho} t^2 f^2 B_m^2 \text{ W/m}^3$$

where  $B_m$  = maximum flux density  
 $f$  = frequency of excitation  
 $t$  = thickness of core  
 $\rho$  = resistivity of the core material

From this expression, it is seen that the eddy-current loss is

- directly proportional to the square of the maximum flux density ;
- directly proportional to the square of the frequency,
- directly proportional to the square of the thickness of the core;
- inversely proportional to the resistivity of the core material.

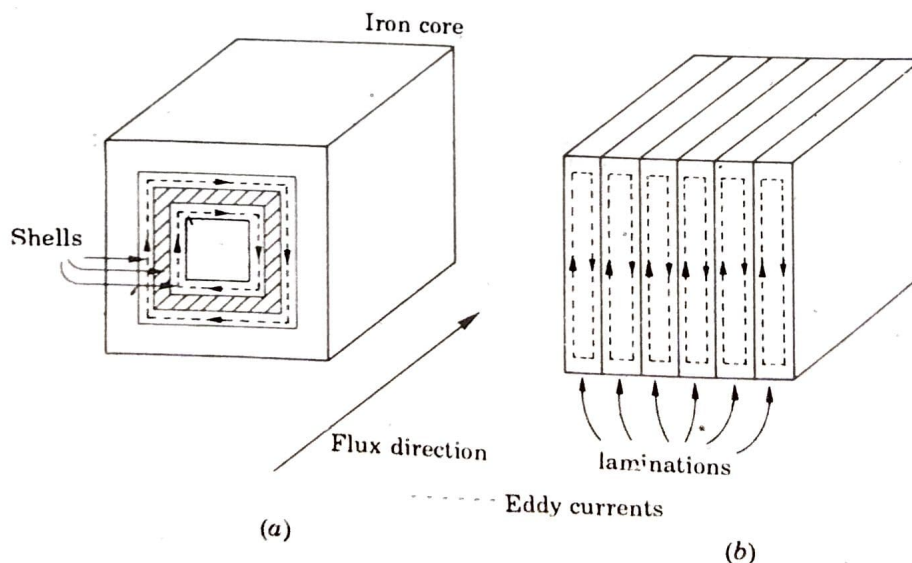


Fig 7.17

### Undesirable effects of eddy currents

Eddy currents are undesirable in inductors, transformer cores and the cores of rotating electric machines as they increase their power losses and produce additional temperature rise.



### Minimising eddy current loss

At low frequencies (50 Hz and audio frequencies) eddy current loss is reduced by making the core of a number of thin sheets, called *laminations*. The laminations (Fig. 7.12 (b)) are lightly insulated from one another by special varnishes. Thus, the closed paths along which eddy currents flow are reduced in length as shown in Fig. 7.12 (b). The lamination thickness varies from 0.5 to 5 mm in electrical machines and from 0.01 to 0.5 mm in devices used in electronic circuits operating at higher frequencies. Since the eddy current loss is proportional to the square of the frequency, this loss becomes excessive at radio frequencies. It is impracticable to make very thin laminations and therefore *granulated* or powdered-iron cores are used. The iron dust is mixed with a binding material and compressed to form the required shape of the core. Thus, particles are insulated from one another and the eddy current path lengths are considerably reduced. Such cores find application at radio frequencies. Eddy currents are further reduced by using the core material of a high resistivity. For this purpose the electrical steel is alloyed with up to 4% of silicon.

### Applications of eddy currents

Some of the applications of eddy currents are as follows :

- (1) Eddy current heating is used for heating metals; for example melting, hardening and other heat-treatment processes.
- (2) Eddy current damping is used in permanent-magnet moving-coil instruments.
- (3) Eddy current braking is used in induction energy meters.

## 121 STACKING FACTOR

Magnetic circuits are usually constructed of laminations which are coated with insulating varnish. Thus, there is a small space present between the successive laminations. Consequently, the effective magnetic cross-sectional area is less than the overall area of the stack. The ratio of the effective area to the overall area is called the *stacking factor*. It is also defined as the ratio of volume occupied by magnetic material to the total volume of the core. This factor is important in calculating flux densities in magnetic parts. Stacking factor is usually less than 1.0. It approaches 1.0 as the lamination thickness increases.

## EXERCISES

1. Define m.m.f., flux, reluctance and permeability.

2. An iron ring of mean length 1.2 m and cross-sectional area  $0.005 \text{ m}^2$  is wound with a coil of 900 turns. If a current of 2 A in the coil produces a flux density in the ring of 1.1 T, calculate (a) the m.m.f.; (b) the total flux in the ring; (c) the magnetic field strength; (d) the relative permeability of the iron at this flux density. [(a) 1800 A; (b) 5.5 mWb; (c) 1500 A/m; (d) 583]

3. An iron ring has a mean length of 1.5 m and a cross-sectional area of  $0.01 \text{ m}^2$ . It has a radial airgap of 4 mm. The ring is uniformly wound with 250 turns. What direct current would be needed in the coil to produce a flux of 0.8 mWb in the airgap? Assume relative permeability of iron as 400 and leakage factor as 1.25. [2.21 A]

4. An iron magnetic circuit has a uniform cross-sectional area of  $5 \text{ cm}^2$  and a length of 25 cm. A coil of 120 turns is wound uniformly over the magnetic circuit. When the current in the coil is 1.5 A, the total flux is 0.3 mWb; when the current is 5 A, the total flux is 0.6 mWb. For each value of current, calculate (a) the magnetising force, (b) the relative permeability of the iron. (U.E.I.) [720 A/m; 663; 2400 A/m, 398]

5. An iron ring, having a mean diameter of 75 cm and a cross-sectional area of  $5 \text{ cm}^2$ , is wound with a magnetising coil of 120 turns. Using the following data, calculate the current required to set up a magnetic flux of 630  $\mu\text{Wb}$  in the ring.